

João José

Academic Curriculum Vitae

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[Academic profile](#) | [LinkedIn](#) | [GitHub](#)

Research profile

Software engineer with research experience at the University of Algarve's Vision Laboratory (2008-2013). Academic work focused on computer vision for the mobility of blind and visually impaired people: real-time path and obstacle detection, indoor landmarks, lateral-door detection and sound feedback. Contributor to SmartVision and Blavigator; author or co-author of a book, journal articles, conference papers and a conference abstract.

Research experience

Computer Vision Researcher / Research Fellow

Vision Laboratory, University of Algarve | 2008-2013 | Faro, Portugal

Developed human-vision models and active-vision algorithms in C/C++ and Matlab. Research member of ISR-Lisbon/IST; collaborated with the University of Trás-os-Montes and Alto Douro (UTAD).

SmartVision. Contributed to a portable navigation aid combining computer vision, geographic information and localisation. The vision work addressed walkable paths and static or moving obstacles beyond the reach of the white cane.

Blavigator. Worked on visual navigation covering path following, obstacle detection and indoor landmarks, including lateral-door detection and sound feedback for navigation guidance.

Education

Doctoral studies in Computer Science Engineering (not completed)

University of Algarve | Sep 2011 - Oct 2013

Robust visual recognition of objects and environments for blind users: literature review and algorithm comparison, an initial object-recognition model, and hand and arm segmentation. Completed the programme expected for the first two years; studies subsequently suspended.

MSc in Computer Science Engineering

University of Algarve | Sep 2008 - Jun 2011 | Degree grade: 16/20

Dissertation: *Real-time path and obstacle detection for blind persons* (2010).

Real-time path and obstacle detection on a small netbook using C/C++, OpenCV and a Qt interface. Supervisor: Prof. Johannes Martinus Hubertina du Buf. Dissertation grade: 18/20.

[Dissertation in the University of Algarve repository](#)

BSc in Computer Science Engineering

University of Algarve | Sep 2003 - Jul 2008

Selected project: a web content-management system for football news and results, using HTML, PHP, JavaScript, PostgreSQL and PL/pgSQL. Project grade: 19/20.

Studies in Electronics Engineering

University of Algarve | Sep 2001 - Jul 2003

Changed major to Computer Science Engineering; no degree awarded in Electronics Engineering.

Publications

Journal articles

- [1] M. Serrão, S. Shahrabadi, M. Moreno, **J. T. José**, J. I. Rodrigues, J. M. F. Rodrigues & J. M. H. du Buf (2015). **Computer vision and GIS for the navigation of blind persons in buildings**. Universal Access in the Information Society, 14, 67-80. Published online in 2014. [DOI](#) | [Open PDF](#)
- [2] **João José**, Miguel Farrajota, João M. F. Rodrigues & J. M. Hans du Buf (2011). **The SmartVision local navigation aid for blind and visually impaired persons**. International Journal of Digital Content Technology and its Applications, 5(5), 362-375. [DOI](#) | [Open PDF](#)
- [3] J. M. Hans du Buf, João Barroso, João M. F. Rodrigues, Hugo Paredes, Miguel Farrajota, Hugo Fernandes, **João José**, Victor Teixeira & Mário Saleiro (2011). **The SmartVision Navigation Prototype for Blind Users**. International Journal of Digital Content Technology and its Applications, 5(5), 351-361. [DOI](#) | [Open PDF](#)

Conference papers

- [4] M. Moreno, S. Shahrabadi, **J. José**, J. M. H. du Buf & J. M. F. Rodrigues (2012). **Realtime Local Navigation for the Blind: Detection of Lateral Doors and Sound Interface**. DSAI 2012 · Procedia Computer Science, 14, 74-82. [DOI](#) | [Open PDF](#)
- [5] **J. José**, J. M. H. du Buf & J. M. F. Rodrigues (2012). **Visual Navigation for the Blind - Path and Obstacle Detection**. Proceedings of the 1st International Conference on Pattern Recognition Applications and Methods (ICPRAM), 515-519. [DOI](#) | [Open PDF](#)
- [6] **J. José**, M. Farrajota, J. M. F. Rodrigues & J. M. H. du Buf (2010). **A vision system for detecting paths and moving obstacles for the blind**. Proceedings of the 3rd International Conference on Software Development for Enhancing Accessibility and Fighting Info-exclusion (DSAI), Oxford, 175-182. [Open PDF](#)
- [7] J. M. H. du Buf, J. Barroso, J. M. F. Rodrigues, H. Paredes, M. Farrajota, H. Fernandes, **J. José**, V. Teixeira & M. Saleiro (2010). **The SmartVision Navigation Prototype for the Blind**. Proceedings of the 3rd International Conference on Software Development for Enhancing Accessibility and Fighting Info-exclusion (DSAI), Oxford, 167-174. [Open PDF](#)

Conference abstract

- [8] **J. Jose**, M. Moreno, J. Pinilla-Dutoit, J. Rodrigues & J. Du Buf (2012). **Blavigator: a navigation aid for blind persons**. European Conference on Visual Perception (ECVP 2012). Perception, 41(1_suppl), p. 99, abstract 97. [ECVP 2012 abstracts · PDF, page 99](#)

Book

- [9] **João Tiago Pereira Nunes José** (2013). **Real-time path and obstacle detection for blind persons**. ISBN 978-3-659-45730-2.

Research grants and awards

SmartVision research grant

Project PTDC/EIA/73633/2006

SmartVision: active vision for the blind.

Blavigator research grant

Project RIPD/ADA/109690/2009

Blavigator: a cheap and reliable navigation aid for the blind.

Doctoral research grant

SFRH/BD/82310/2011

Robust visual recognition of objects and environments for blind persons.

CERATONIA award grant

3.70.13.001

The Amateur Painter.

Teaching

Cisco CCNA Exploration Instructor

Cisco Academy, FCT - University of Algarve | Nov 2008 - Jun 2009 | Faro, Portugal

Network Fundamentals. Network models and protocols, Ethernet, IP addressing and subnetting.

Routing Protocols and Concepts. Routing fundamentals, VLSM, RIP, OSPF and EIGRP.

LAN Switching and Wireless. Switch configuration, VLANs, trunking, spanning tree and WLANs.

Accessing the WAN. WAN technologies, ACLs, PPP/PPPoE, Frame Relay and VPN concepts.

Training and qualifications

Certificate of Pedagogical Competences (CCP)

IEFP | Mar 2015

Initial trainer training at Training House, Faro (Jan-Mar 2015), covering pedagogical methods and competences.

Cisco Certified Network Associate (CCNA)

Cisco Academy, FCT - University of Algarve | Oct 2007 - May 2008

Modules 1-4; average grade: 90%.

Professional Scrum Master I

Scrum.org | Mar 2018

Languages

Portuguese (native), English (fluent), Dutch (intermediate), Spanish (good).

Selected professional and applied experience

Independent Software Engineer at JTJ Software

Nov 2022 - Present | Netherlands and Portugal

LVNL SIM Builder (Jun 2024 - Feb 2026). Contributed across the full stack of a web-based ATM scenario platform built on an adapted BlueSky simulator, combining development and Scrum Master responsibilities.

LVNL iCAS SIM (Nov 2022 - Dec 2025). Developed and maintained the simulation and training environment for the Netherlands' next-generation air traffic management system, including integration with peripheral systems.

Portuguese real-estate software (Jan 2026 - Present). Product discovery, architecture and implementation for the Portuguese real-estate market.

C++ Software Developer at Accent Pointe B.V.

May 2021 - Nov 2022 | Netherlands

Developed trading software integrated with brokers and market-data feeds; modernised a legacy MFC and Microsoft RPC codebase.

C++ Software Developer at ABN AMRO Clearing Bank

Nov 2016 - May 2021 | Netherlands

Delivered financial risk-management software and integrated service interfaces, messaging and authentication components.

Software Engineer at Frinch

Apr 2016 - Oct 2016 | Faro, Portugal / Remote

Developed Product Information Management software with a distributed team working with the company's Munich headquarters.

Software Engineer and Lead Researcher at Evision Industry Software

Oct 2013 - Sep 2015 | Faro, Portugal

Developed algorithms for text detection, raster-to-vector conversion and automatic interpretation of piping and instrumentation diagrams and AutoCAD files, including pipe and instrument recognition. Worked with the team in The Hague using C# and TypeScript.

Invited Expert at SEF

May 2007 - Aug 2007 | Faro, Portugal

Participated in the quality assessment of the electronic border-control system installed at Faro Airport.

Technical skills

Research and computer vision. C/C++, Matlab, OpenCV and Qt; real-time path and obstacle detection, visual object recognition, text detection and raster-to-vector conversion.

Software engineering. C++, Python, C, Go, Ada, Bash, C# and TypeScript; HTML, PHP and JavaScript; REST APIs, ZeroMQ, Kafka and Protobuf; PostgreSQL, PL/pgSQL and Ingres; Git, CMake, GCC and GDB; pytest and C++ testing tools; Linux, Docker and OpenShift.